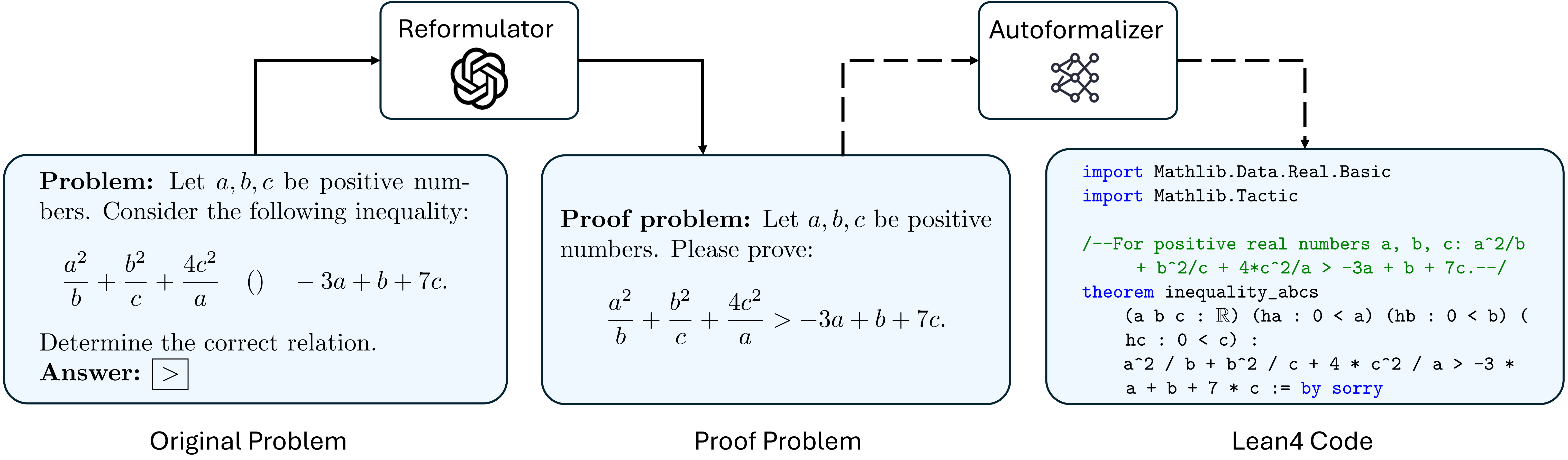

Reformulator
Autoformalizer
Problem: Let a, b, c be positive numbers. Consider the following inequality:
$\frac{a^2}{b} + \frac{b^2}{c} + \frac{4c^2}{a} \quad () \quad -3a + b + 7c.$
Determine the correct relation.
Answer: $\boxed{>}$
Proof problem: Let a, b, c be positive numbers. Please prove:
$\frac{a^2}{b} + \frac{b^2}{c} + \frac{4c^2}{a} > -3a + b + 7c.$
import Mathlib.Data.Real.Basic
import Mathlib.Tactic
/--For positive real numbers a, b, c: a^2/b + b^2/c + 4*c^2/a > -3a + b + 7c.--/
theorem inequality_abcs
(a b c : ℝ) (ha : 0 < a) (hb : 0 < b) (
hc : 0 < c) :
a^2 / b + b^2 / c + 4 * c^2 / a > -3 *
a + b + 7 * c := by sorry
Original Problem
Proof Problem
Lean4 Code